
Single night in 2024
15,401 homeless
9 km
on average to travel
0.55%

Minimizing Travel Distance: A Person-Centred Modeling Framework for Homeless Shelter Accessibility in Toronto

DEFY GRAVITY

Group 7

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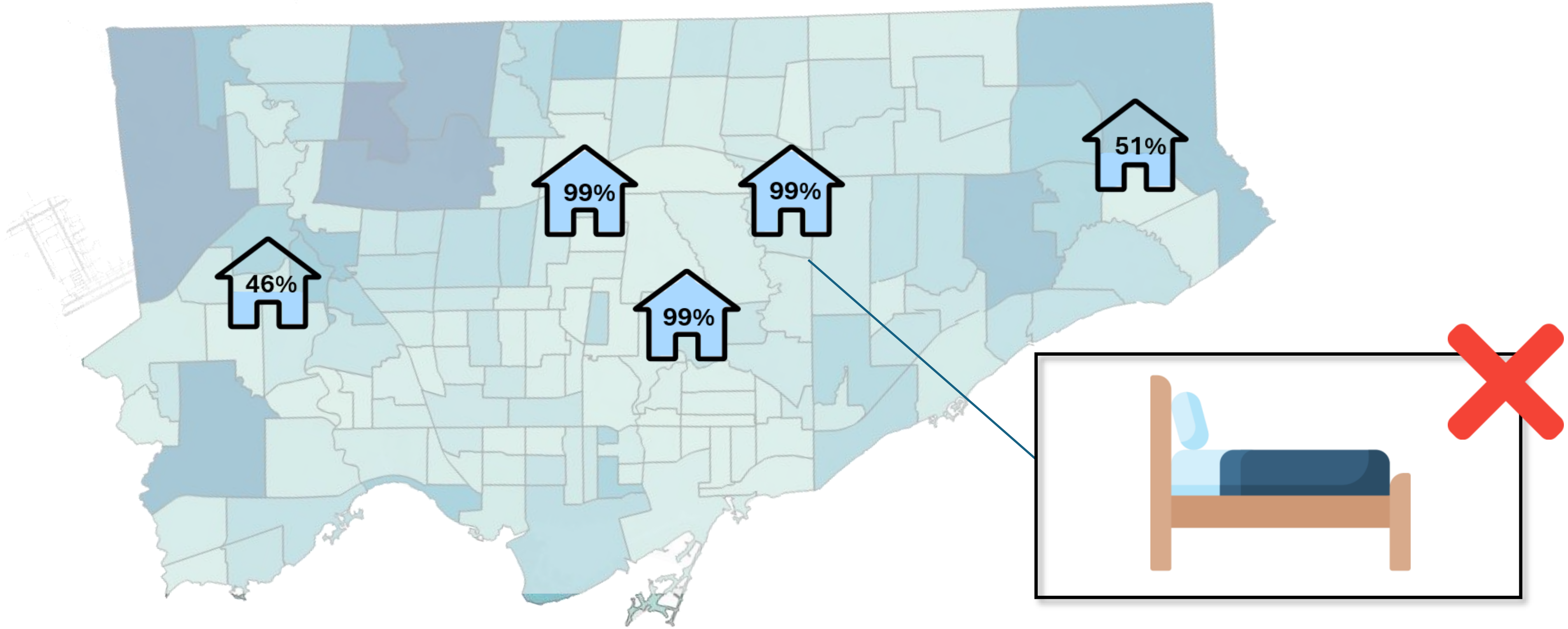
Problem Analysis

Background

Homelessness in Toronto often means **travelling across the city in search of a bed that may not be available.**

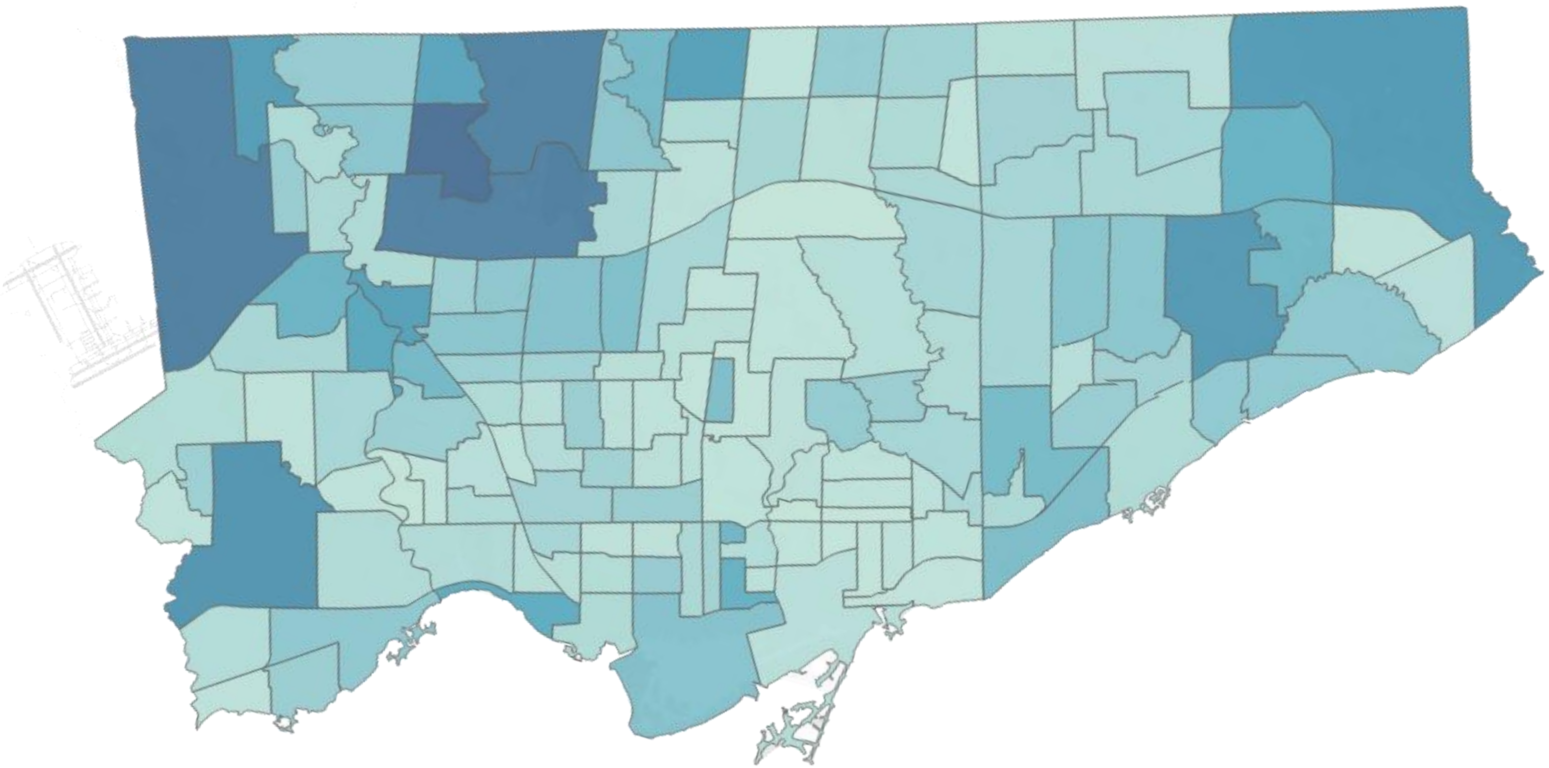


Background Data Methods & Results Conclusion

Spatial Mismatch

Background

High-demand areas often have few nearby shelters, while some shelters in lower-demand areas still have available beds.



Main Problem

Background

Problem Statement

"How can we **allocate shelter spaces** to **minimize travel distance** while respecting capacity, eligibility, and geographic imbalances?"

Our Approach

1
Identify Where Demand Comes From (Hotspots)

2
Map Travel Distances to Shelter Network

3
Allocate Strategically using Optimization & Simulation Framework

Data Sources

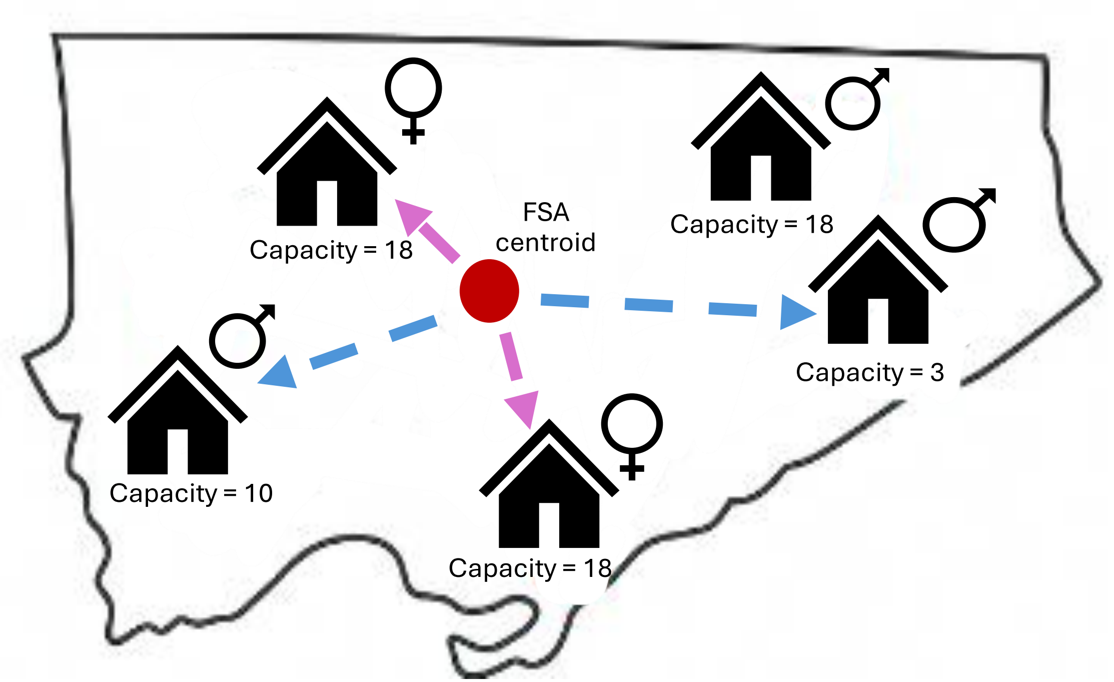
Data

Source	Dataset	Purpose
OPEN DATA	Monthly Shelter Flow 2021-2025, 717 rows x 20 columns	Identify demand patterns
	Shelter Occupancy & Capacity 2024, 54,000 rows x 14 columns	Track capacity & flows
Statistics Canada / Statistique Canada	2021 Census by FSA 2021, 3 rows x 98 columns	Define high-risk neighbourhoods

Assumptions

Data

1. Centroid of each FSA is **origin of all homeless individuals** (hotspot)
2. Population is restricted to **single adults** who are **male and female**
3. Capacities are **fixed** for all months
4. Travel distances are **straight line** from centroid to FSA
5. All individuals **restart at the origin every monthly assignment time**



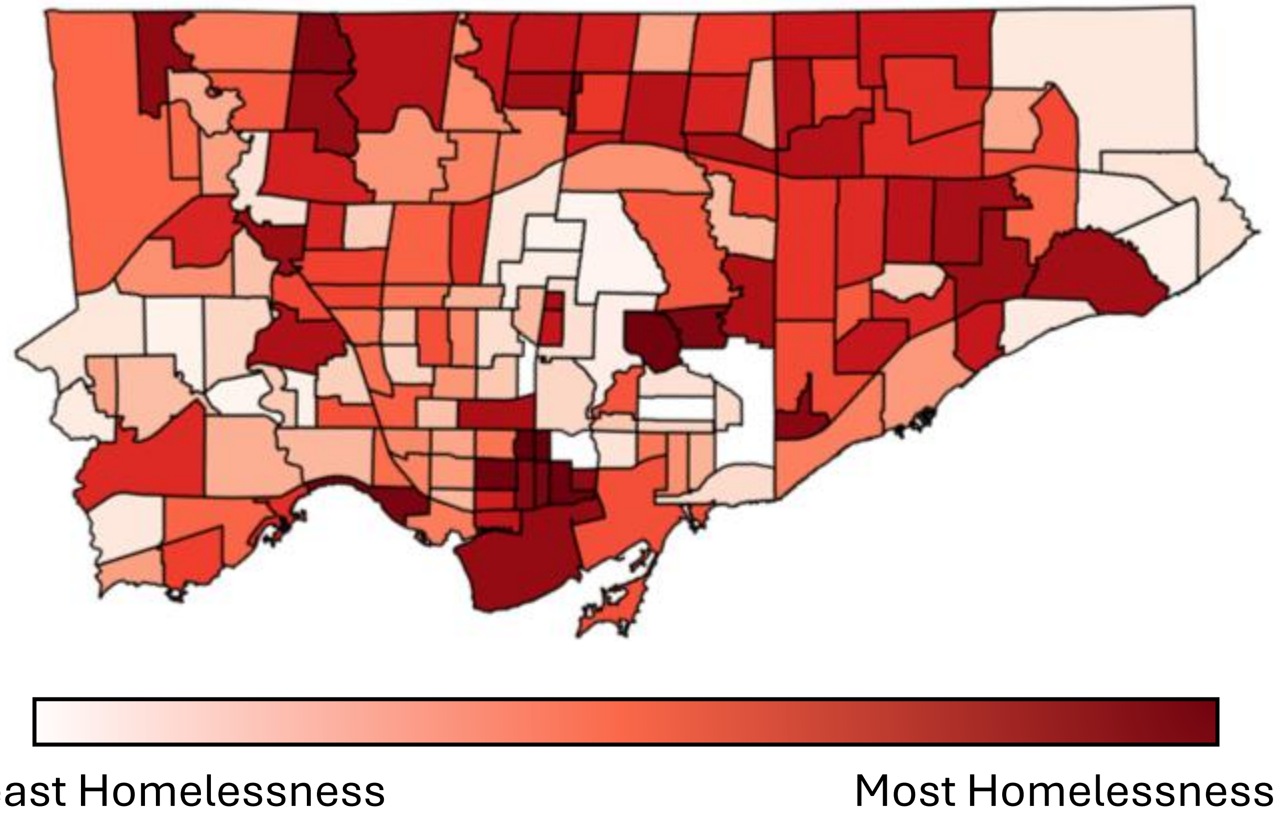
Hotspots

Data

Socioeconomic indicators

- low income
- housing-cost burden
- core housing need

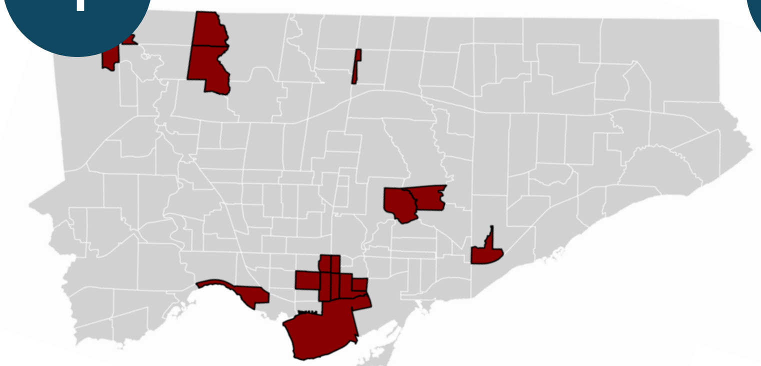
help identify Toronto FSA's where homelessness pressures are highest.



Distances

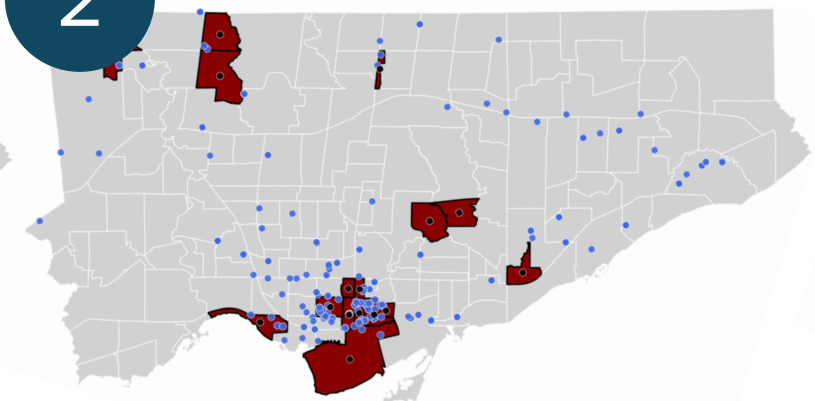
Data

1



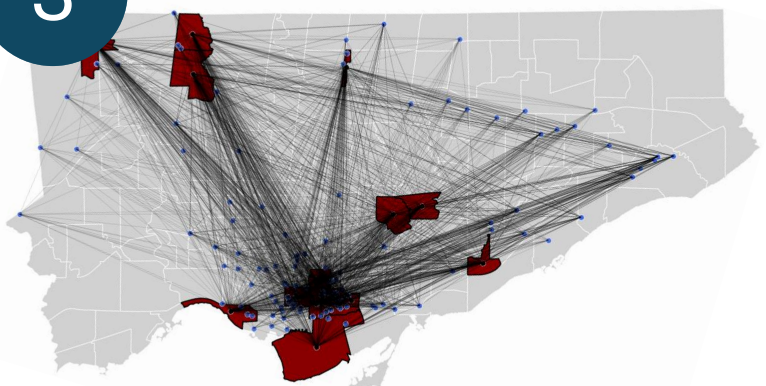
Isolate Top
16 FSAs

2



Geocode
Centroid & Shelters

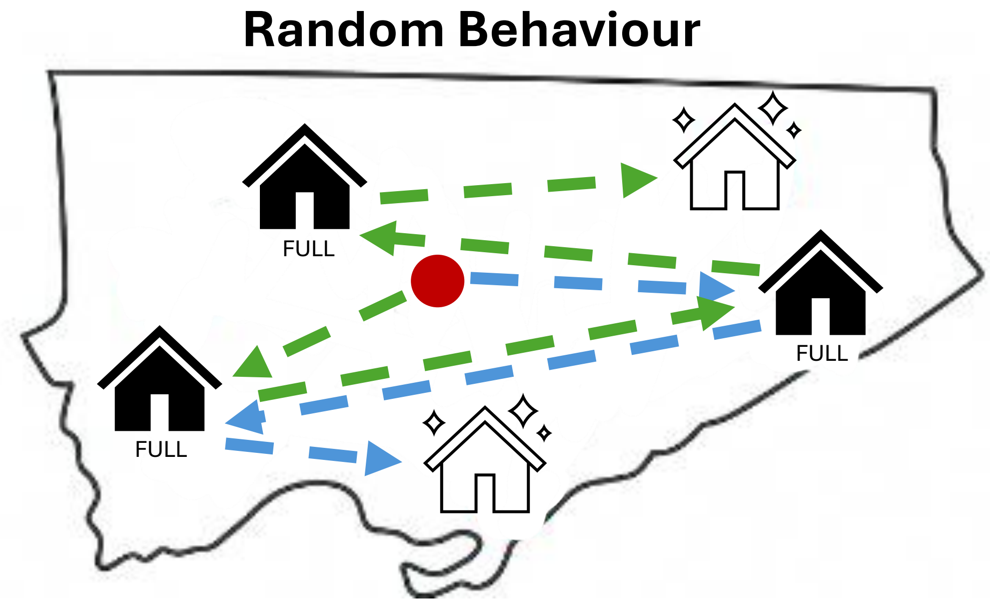
3



Compute Travel
Distances Matrix

Simulation

Methods & Results



How does it work?

Our simulation models how people might actually seek shelter: either **choosing randomly** or going to the **closest shelter they believe has space**.

Simulation

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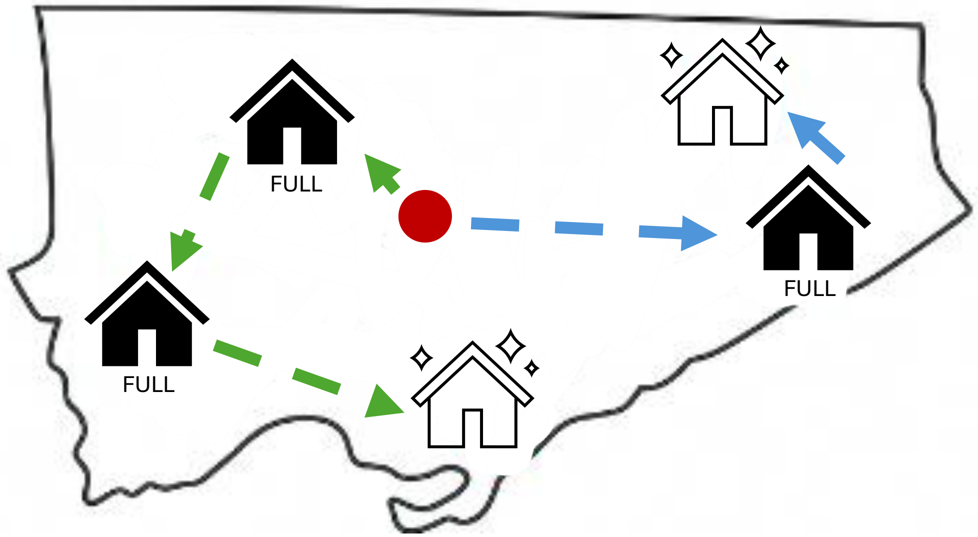
Logical (Greedy) Behaviour



Baseline System

Methods & Results

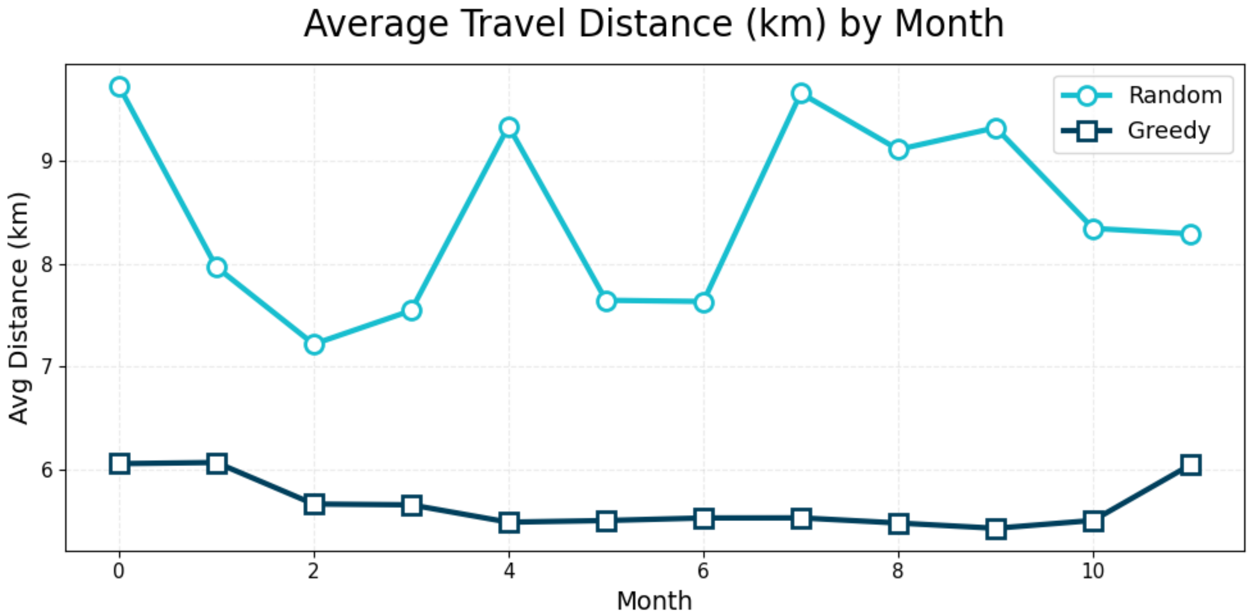
Average Travel Distance (km) by Month

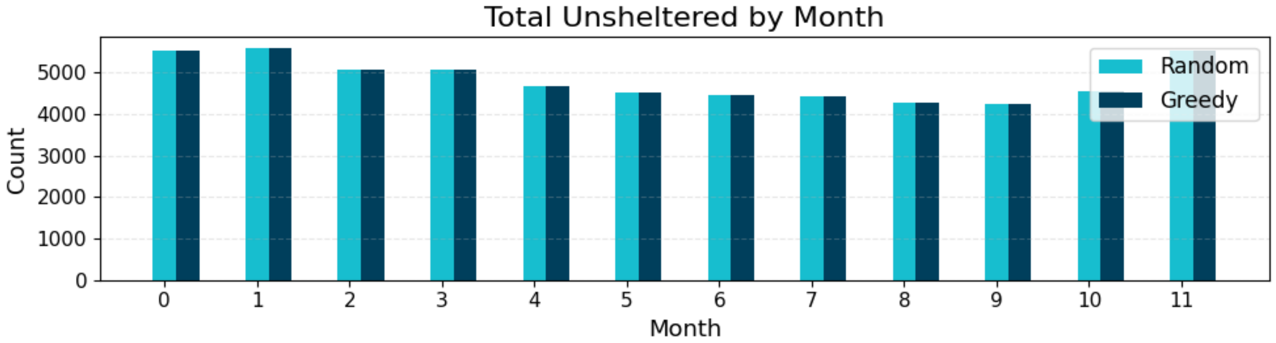
~9 km
Random Behaviour

~6 km
Greedy Behaviour

Total Unsheltered by Month

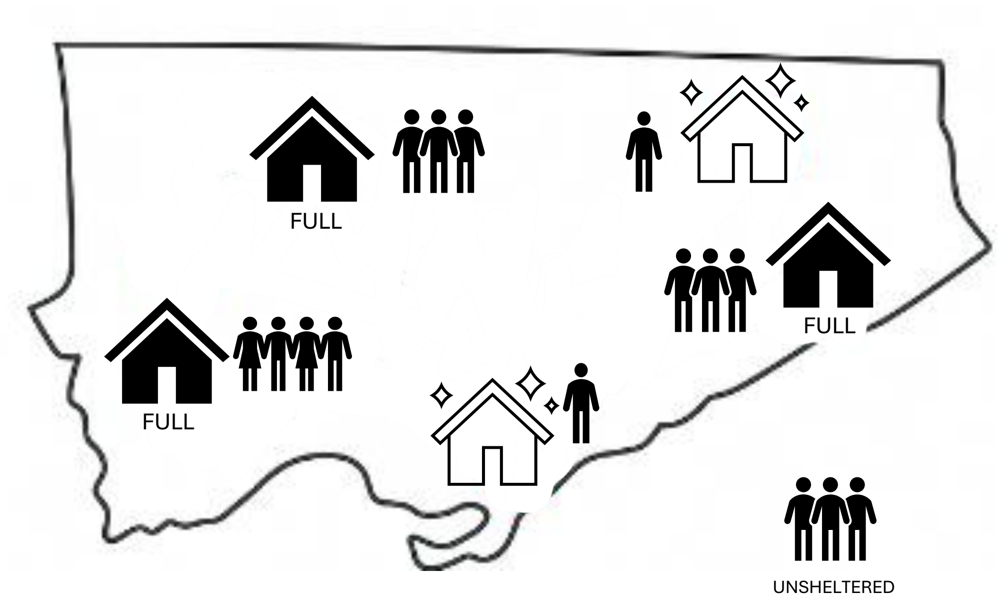
100%
Capacity





Optimization – Stage 1

Methods & Results



How does it work?

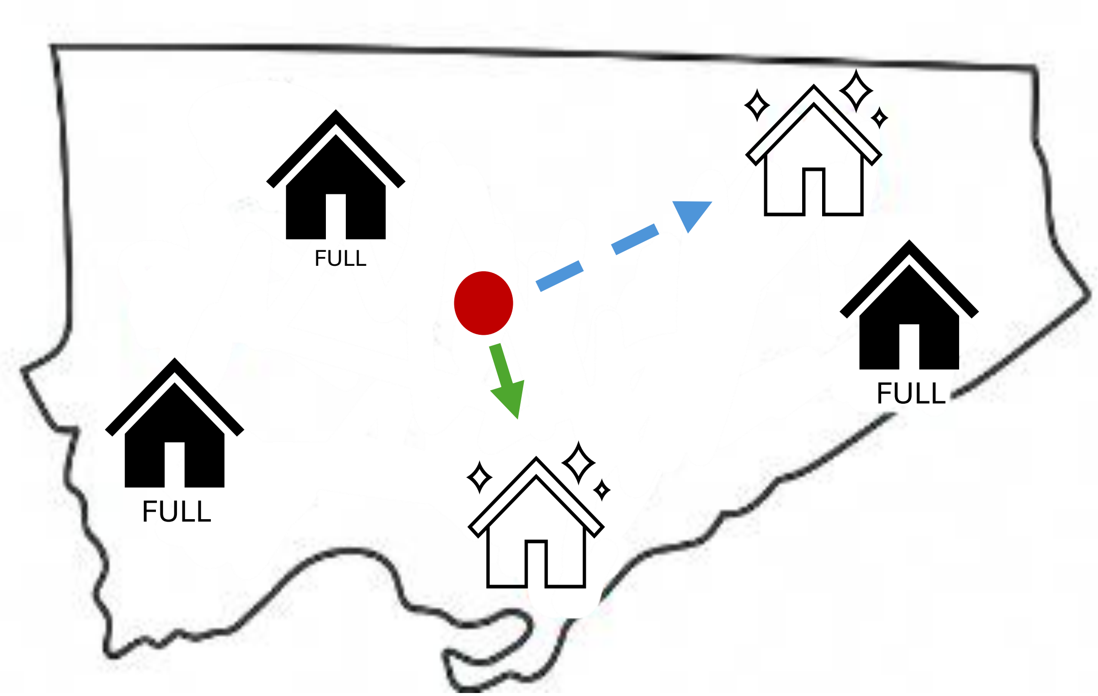
Stage 1 assigns **as many individuals as possible** to shelters while considering **capacity limits** and **gender eligibility**, minimizing the number left unsheltered.

Optimization – Stage 2

Methods & Results

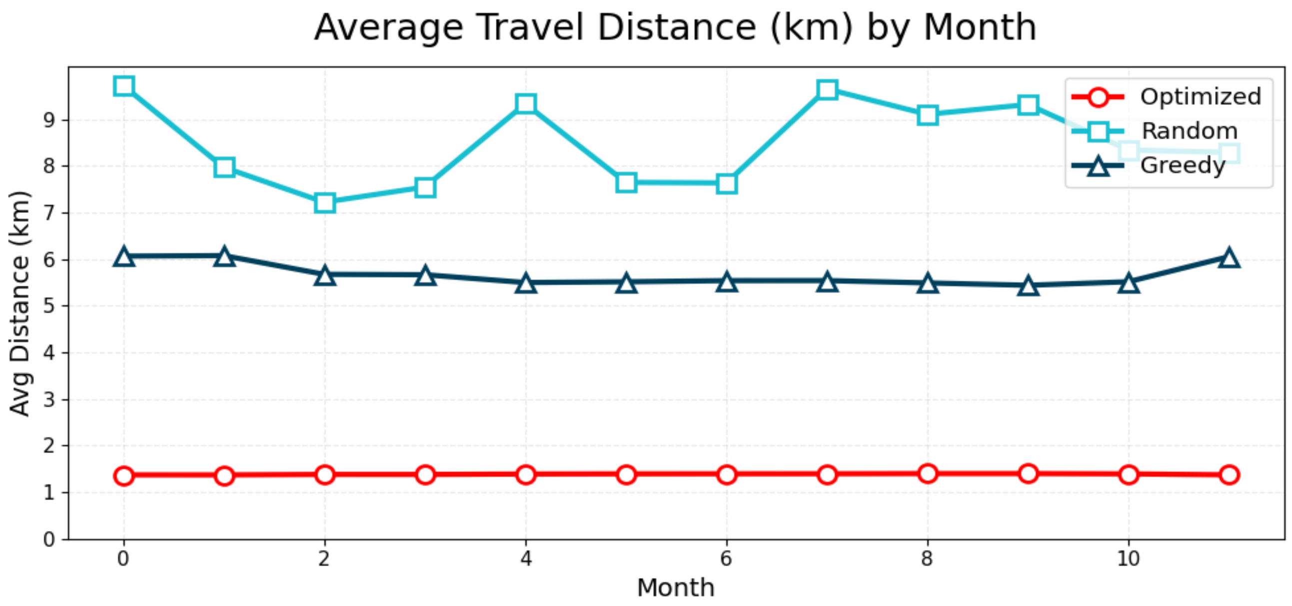
How does it work?

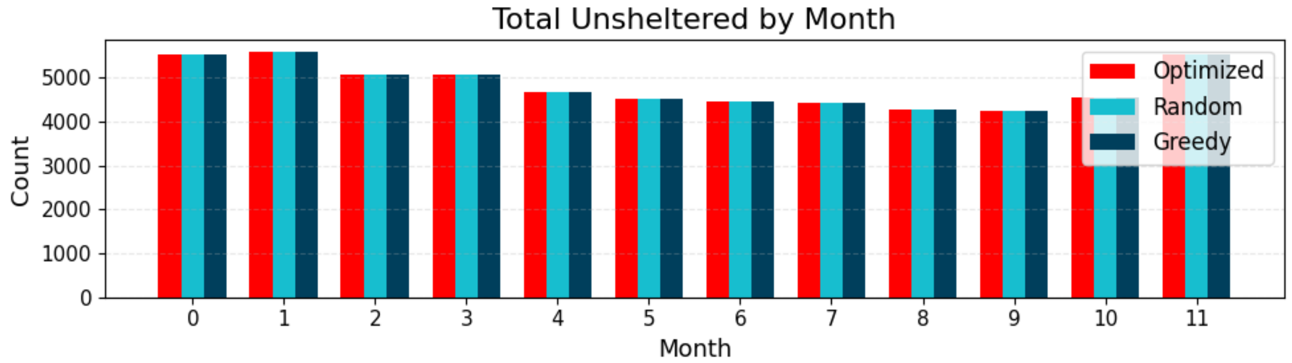
Stage 2 then finds the **arrangement of those placements** that produces the **lowest total travel distance** across all individuals.



Optimized System

Methods & Results





Average Travel Distance (km) by Month

~1.3 km

Distances

Total Unsheltered by Month

100%

Capacity

Conclusion

Conclusion

Simulation

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Optimization

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